

RALLY GPS

Supplementary live tracking and stage-flag safety tool
for the Cyprus Rally organiser and race control.

PREPARED FOR

FIA/NATIONAL — event safety /
operations review

EVENT

Cyprus Rally

DOCUMENT DATE

September 2026

STATUS

Software application briefing

Purpose and scope

WHAT RALLY GPS IS

- Live GPS from competing crews' smartphones to race control.
- Authenticated HQ map, plus Google Earth Pro live KML.
- Per-event KMZ: road, special stages, and control pins.
- GREEN FLAG / RED FLAG for cars currently on that special stage.
- Stopped-crew OK / SOS confirmation on stage.

System overview

1 DRIVER DEVICE

- Android application or HTTPS web page.
- Sends GPS and crew actions (OK / SOS / flag ack).

2 RALLY GPS SERVER

- Receives live fixes and offline batches.
- Binds each car to the LIVE rally KMZ.
- Stores flags and SOS.

3 RACE CONTROL HQ

- Authenticated dashboard.
- Full-screen map beside sidebar.
- Cars as numbers; stage flags.

4 GOOGLE EARTH PRO

- Network Link KML.
- Refresh every 2 seconds.
- Same live rally feed.

REFRESH AND LIVE STATE

- HQ map polls cars about every 2 seconds. Driver devices poll stage-flag state about every 2 seconds (web) or 2 seconds (Android).
- A car is LIVE while tracking and a fix has been received within 3 minutes; otherwise HQ shows last GPS / no signal.

Rally and route binding

NAMED EVENT

- HQ creates a rally with name and dates.
- Cyprus Rally is a distinct event, not a global route.

LIFECYCLE

- draft → LIVE → ended.
- Cars are drawn on the map only while a rally is LIVE.

ONE LIVE RALLY

- Going live snapshots and ends any other live rally.
- HQ is bound to a single course.

PER-RALLY KMZ

- KMZ/KML is uploaded for the selected rally only.
- Other events keep their own roads, stages, and pins.

GO LIVE AGAIN

- An ended rally can be set live again to review the stored snapshot.

CLEAR / REPLACE

- HQ can clear or replace that rally's route without affecting other events.

Live tracking and car display

RACE CONTROL MAP

- Cars appear as competition numbers (no live trail clutter).
- Click a number: name, motion, speed (km/h), section, OK/SOS.
- Green = moving (~4 km/h+); blue = stopped; red = SOS.
- List: LIVE · NO SIGNAL / LAST GPS · STOPPED.
- Trails are not drawn live by default; HQ may show one route or download GPX.
- “Refresh lost cars” holds last GPS and requests reconnect.

MARKER COLOUR

- Green — moving (speed above ~1.2 m/s / ~4 km/h).
- Blue — on course but not moving.
- Red — crew has declared RED SOS.

GOOGLE EARTH PRO

- HQ downloads a session-bound Network Link KML.
- Earth refreshes every 2 seconds. Route colours: blue road, red stage.

Stages, road sections, and time controls

KMZ COLOUR RULE

- Red line = special stage.
- Blue line = road / liaison.
- HQ can reclassify a path after upload.

PLACEMARKS

- TC, Start, Finish, Stop, Refueling.
- Google Earth-style icons on the map and live KML.
- Optional PNG replace per icon type, per rally.

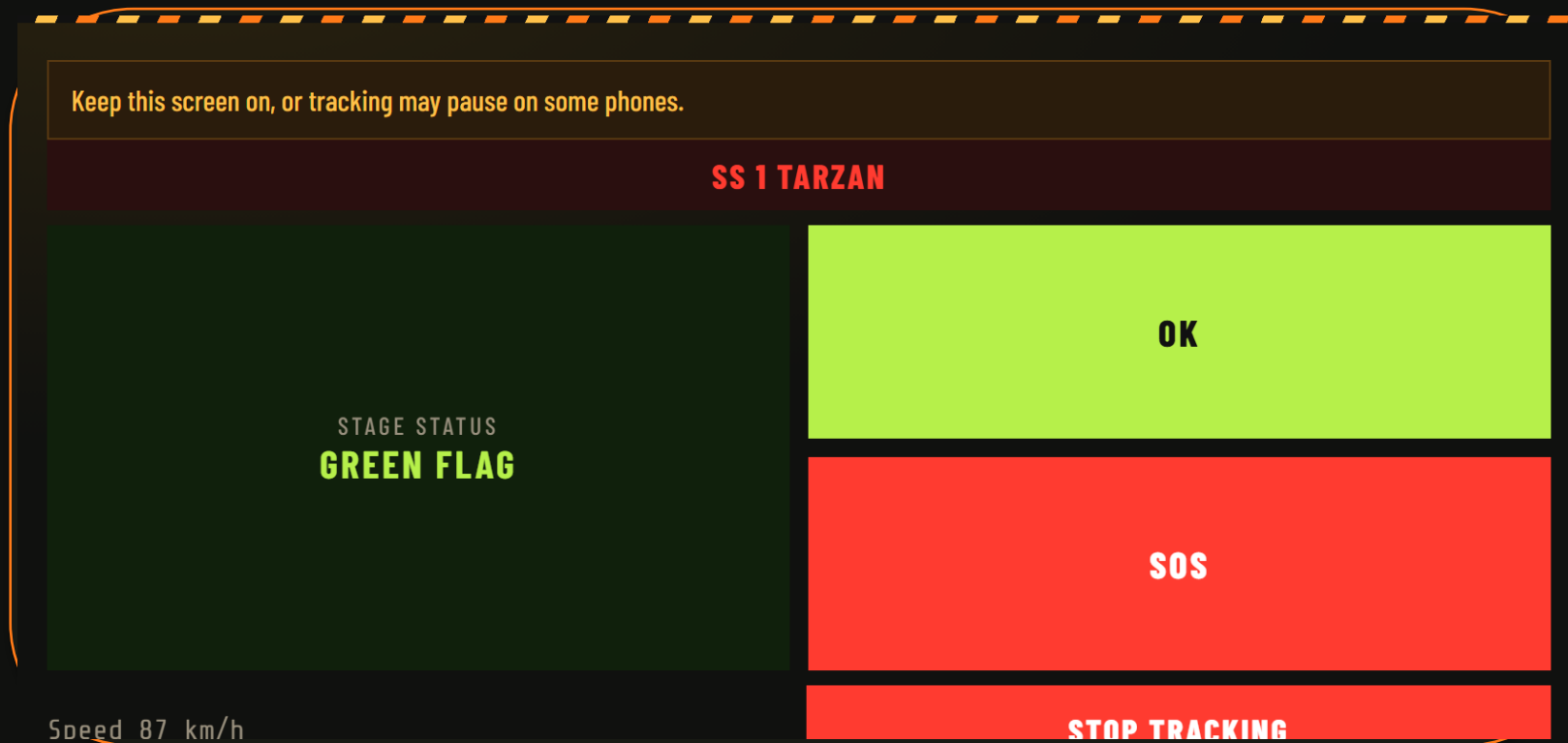
HQ TABS

- All · Stages · Road · Time controls.
- Counts shown per tab.
- Click a section to fit it on the map.

HOW A CAR IS BOUND TO A SECTION

- The server matches the latest GPS fix to the LIVE rally KMZ.
- Special stage if within about 80 m of a stage line (or inside a stage polygon); road if within about 120 m. If both are close, the stage wins.
- The driver screen shows that KMZ stage name and switches to stage mode. This is KMZ proximity, not an official start/finish beam.

Stage mode — GREEN FLAG



ON-STAGE DISPLAY

- Stage name from the event KMZ.
- STAGE STATUS: GREEN FLAG while that stage is not red-flagged.
- Large OK and SOS controls.
- Speed readout.
- Stop tracking remains available.
- Web driver: keep the screen on unless the Android application is used.

RED FLAG procedure

1 HQ

- On the Stages list, race control sets Red flag for that special stage only. Roads cannot be red-flagged.

2 SERVER

- The flag and a timestamp are stored on that stage. Only cars currently detected on that stage inherit RED FLAG.

3 CARS

- Devices receive flag state on GPS ping and on poll (web ~2 s, Android ~2 s) when connected.

4 CREW

- Those cars get a full-screen RED FLAG, looping alarm, and must tap OK / CONFIRMED.

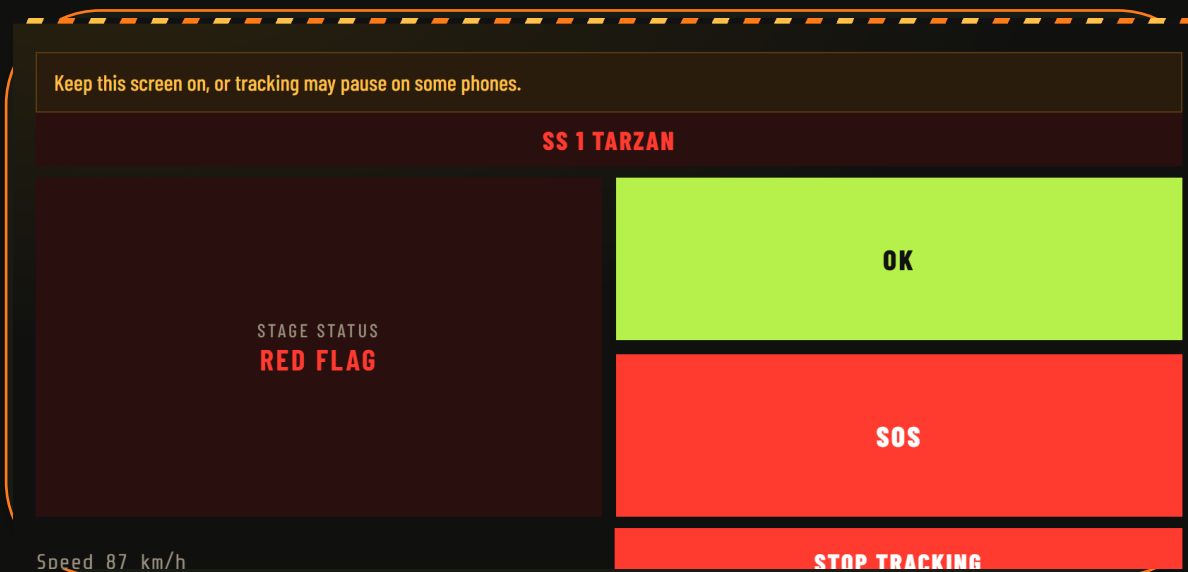
5 HQ

- The car list shows RED FLAG WAITING until acknowledgement, then RED FLAG ACKED.

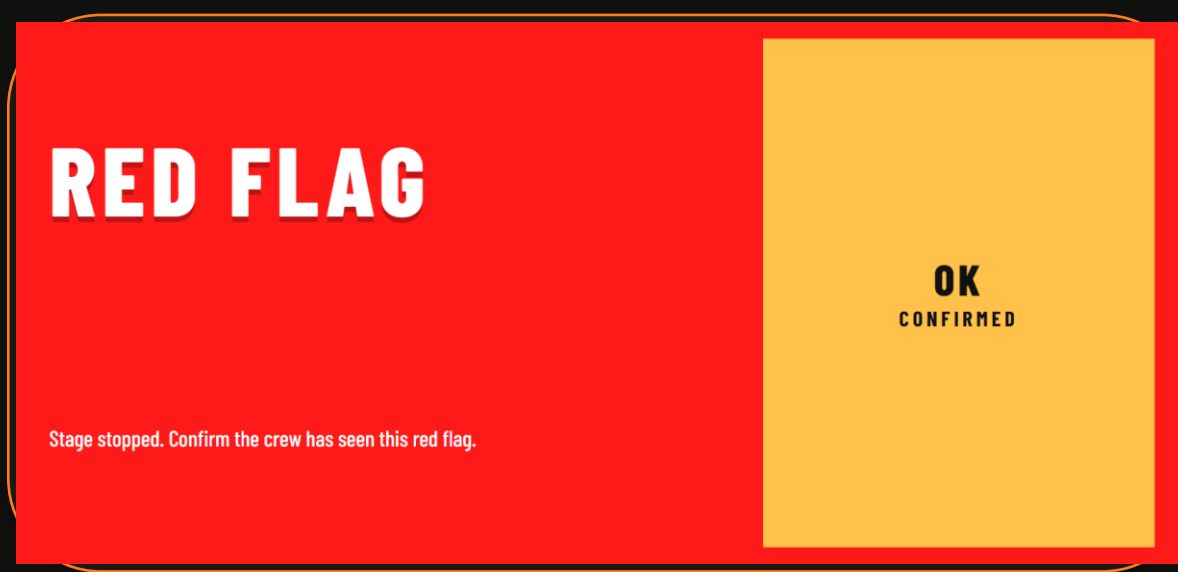
6 CLEAR

- Green flag returns the stage to normal. A later red flag is a new event (new timestamp).

RED FLAG — display and acknowledgement



Stage panel after HQ red-flags that stage.



Full-screen alert until OK / CONFIRMED.

CREW-FACING BEHAVIOUR (AS IMPLEMENTED)

- Full-screen RED FLAG overlay with flashing treatment and a looping alarm. Message: “Stage stopped. Confirm the crew has seen this red flag.”
- Shown only to cars currently ON that red-flagged special stage, until they acknowledge. After ack, the panel stays RED FLAG while HQ keeps the stage red.
- If the device is offline, the flag arrives on the next successful poll/ping; GPS continues to queue locally.

Stopped-crew and SOS procedure

AUTOMATIC PROMPT ON STAGE

- On a special stage, if speed stays below ~4 km/h for more than 20 seconds, STOPPED ON STAGE appears.
- GREEN OK — both crew OK. RED SOS — need help.
- Manual OK / SOS are also available in stage mode.
- An unacknowledged RED FLAG overlay takes priority over this prompt.

WHAT RACE CONTROL SEES

- GREEN OK or RED SOS on the car list.
- SOS forces the map marker red.
- Crew status is tied to the current stage and clears when the car leaves it.
- This does not replace official SOS boards, radio, medical, or rescue.